



Can Drones Be Used To Save A Patient's Life?

This decade has witnessed a great deal about drones and its innumerable purposes. Drones are and can be used in various industries, but there is significant growth of drones in the agricultural sector. Now, this vehicle has been introduced to the medical industry, too



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Currently, there is considerable traffic in cities, especially at peak hours. Imagine that a person has a minor heart attack. What would likely happen next? Let us say people around this emergency spot see the situation and immediately call for an ambulance. The ambulance that comes by road takes minimum time to reach the spot where the emergency has occurred. When there is a patient who needs urgent attention of a doctor, there is a chance that the person might not survive till the time the ambulance arrives. Time taken by the ambulance plays a vital role in deciding the fate of the patient. Thus, it is essential to bring about a distinct means that would take saving human lives one step closer.

This is where drones could serve the purpose. A drone is an aerial vehicle that is not driven by a human. It uses motors and propellers for movement. The more the number of motors and propellers, the more thrust it can create. So, the best choice to use is a quadcopter, as the system comprises four brushless direct current motors (BLDC), which are used along with propellers to give them the necessary thrust.

A 2200mAh electrical storage device can power the drone. The drone well-equipped with a patient monitoring system has been designed to fly to an emergency and extract the victim's real-time health parameters.

The central idea of this is that the doctor sitting in the ambulance, which is still on the way, can see the various results produced by sensors

and observe the patient's condition. Also, once this is done, the doctor can immediately prepare the medical prerequisites that would be required by the patient at the spot of the accident. Moreover, if the condition is critical, the doctor can inform the hospital well in advance so that they can be ready to give immediate medical attention. This not only serves as an excellent medi-aid for the patient but can also result in saving his or her life.

Quadcopter fabrication and design approach

Fabrication of the quadcopter went through a sequence of steps. First, four aluminium rods were cut. Then, the hub was cut, and holes were drilled on aluminium rods, which were measured according to the need of the propeller base. Then, the hub and aluminium rails were integrated to form an x-like structure for the quadcopter frame.

The next step was to incorporate four



Fig. 1: Quadcopter architecture